

Deep inpainting of segmentation gaps in Arabidopsis root phenotyping masks

Gergő Honyák · Group O4 · Breda University of Applied Sciences · Netherlands Plant Eco-phenotyping Centre (NPEC)

Group O4 research focus: Improving automated extraction of plant phenotypic traits from imaging data using computer vision and deep learning.

INTRODUCTION

The problem

Rhizobox imaging projects 3D roots onto a 2D plane through soil. The resulting binary masks contain gaps that break root connectivity and corrupt phenotypic traits such as root length, and branching structure.



Fig. 1a. Gaps caused by soil occlusion — soil blocks parts of the root from the camera view.

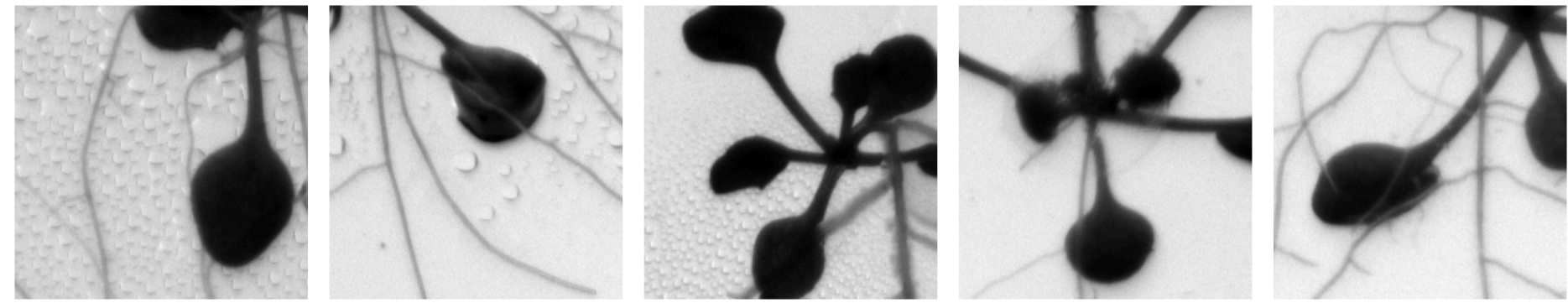


Fig. 1b. Gaps caused by segmentation errors — the model fails to predict root pixels in low-contrast regions.

	H0	Ha
H1	Deep inpainting does not achieve lower MAE within gap regions compared to morphological baseline methods	Deep inpainting achieves lower MAE within gap regions compared to morphological baseline methods
H2	Inpainting does not reduce connected components compared to morphological approaches	Inpainting reduces connected components compared to morphological approaches
H3	Overall mask F1 after deep inpainting does not differ from that achieved by morphological methods	Overall mask F1 after deep inpainting is higher than that achieved by morphological methods

METHOD

Pipeline overview



Fig. 2. DCGAN-generated synthetic root patches (75% real / 25% synthetic per batch).

Two U-Net variants trained and compared:

BCE U-Net binary cross-entropy loss only

BCE+Dice U-Net adds Dice loss for class imbalance

Baseline 1 morphological closing (1×45 kernel)

Baseline 2 graph-based endpoint connection

Metrics computed exclusively within gap regions (Chen et al. 2018): MAE gap, gap F1, connected component delta. Custom GapMetricsCallback ensures honest gap-only evaluation per epoch.

GROUP CONNECTION

Role in group O4

Group O4 investigates how computer vision can automate plant phenotyping pipelines. This work addresses the preprocessing stage: improving mask quality before any trait is computed. More complete masks directly improve accuracy for all downstream trait extraction tasks studied across the group.

62k patches	456 source images	4 methods
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SYNTHETIC GAP GENERATION

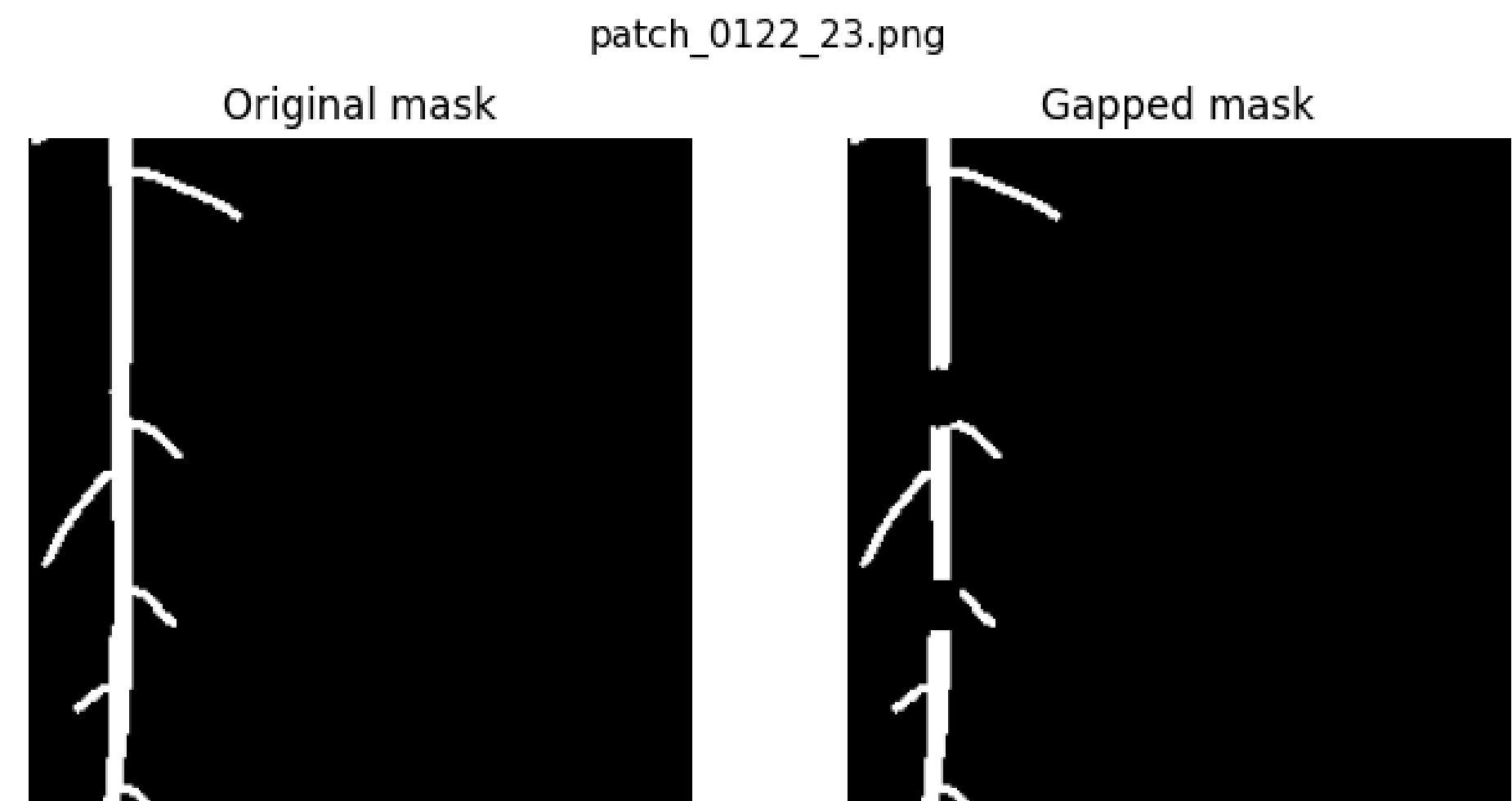


Fig. 3. Synthetic gap generation: original mask (left) vs. artificially gapped version (right) used to create supervised training pairs.

RESULTS

Quantitative comparison across all methods

Method comparison across key gap metrics

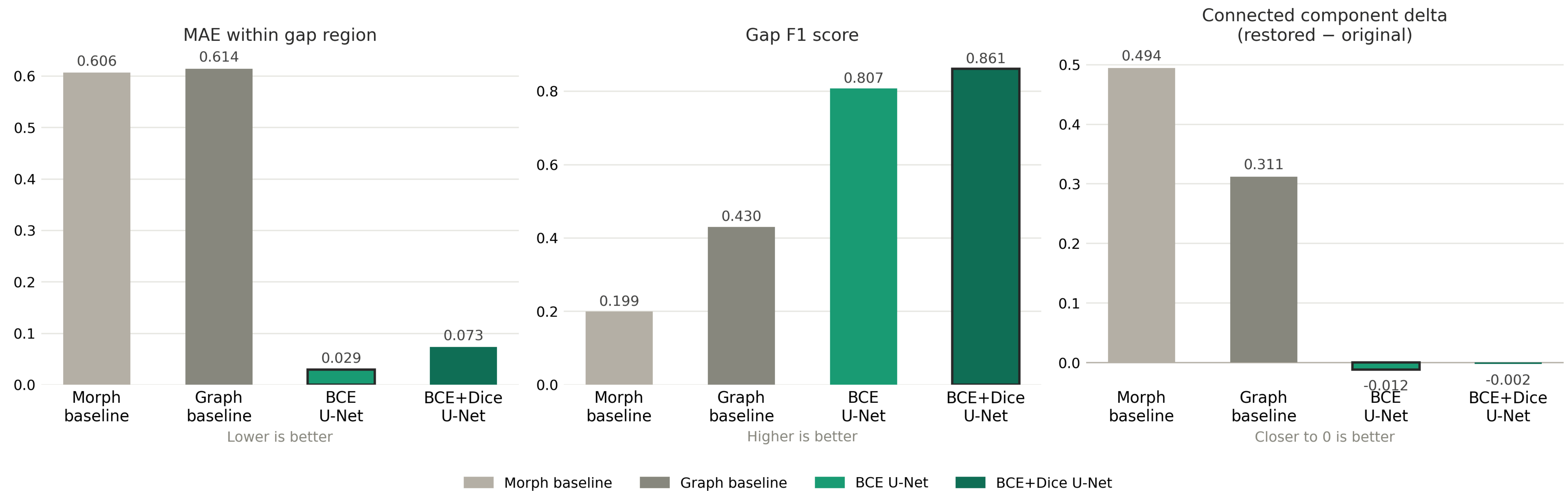


Fig. 4. MAE within gap region (lower is better), gap F1 score (higher is better), and connected component delta (closer to 0 is better). Outlined bars indicate the winner per metric.

RESULTS

Qualitative comparison

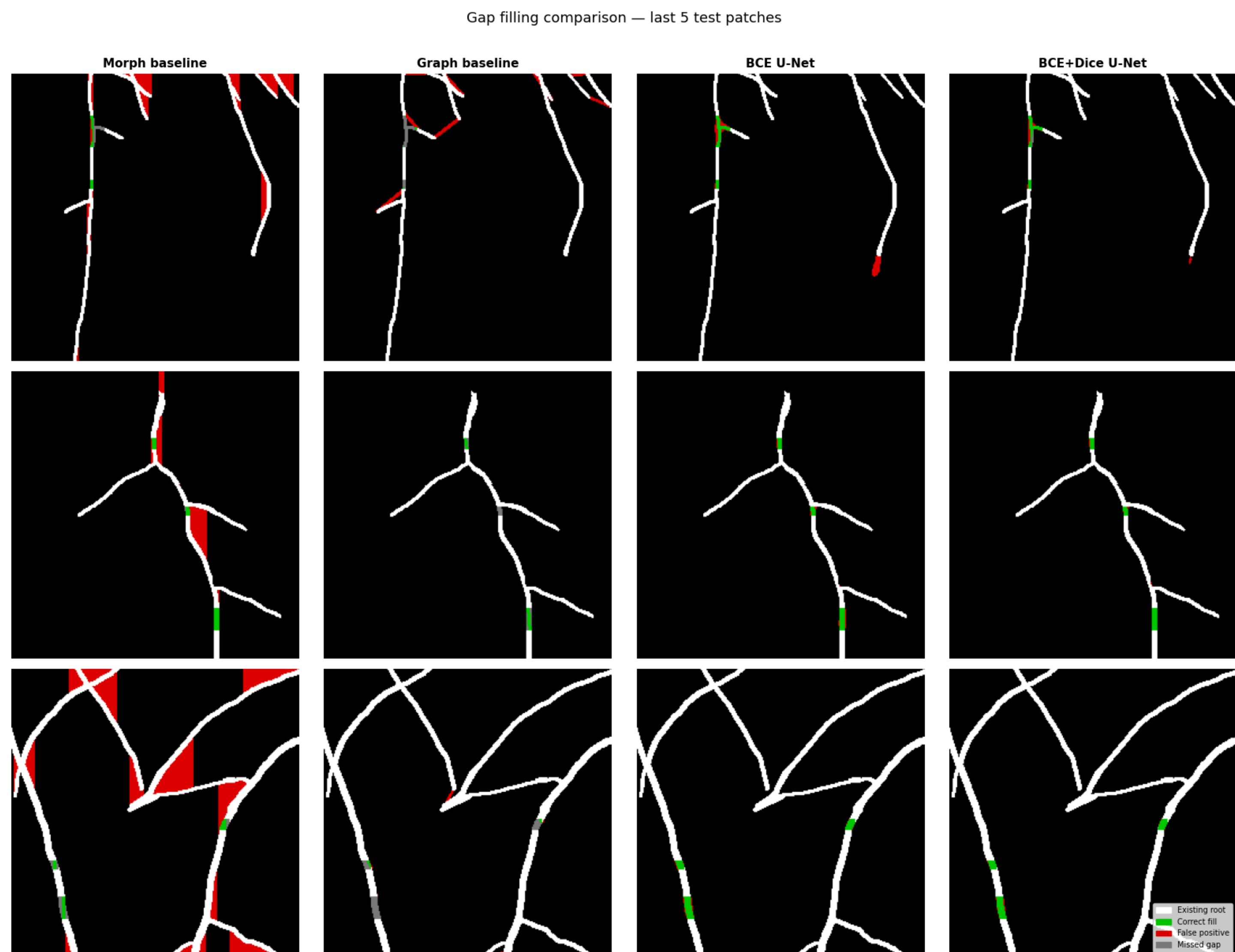


Fig. 5. Gap filling on 5 test patches. Green: correct fill. Red: false positive. Grey: missed gap. Morph baseline introduces large false positive regions. Both U-Net variants fill gaps precisely.

CONCLUSIONS

Hypotheses outcome

Hypothesis	Metric	BCE U-Net	BCE+Dice U-Net	p-value	Result
TESTING BETTER MODEL					
H1	MAE gap	0.029 ✓	0.073	< 0.001	Supported
H2	CC delta	-0.012 ✓	-0.002	< 0.001	Supported
H3	Gap F1	0.807	0.861 ✓	< 0.001	Supported
BENCHMARK VS BASELINES (BCE U-NET)					
H1	MAE gap	0.029 ✓	Morph 0.606 · Graph 0.614	< 0.001	Supported
H2	CC delta	-0.012 ✓	Morph 0.494 · Graph 0.311	< 0.001	Supported
H3	Gap F1	0.807 ✓	Morph 0.199 · Graph 0.430	< 0.001	Supported

BCE U-Net is selected as the representative deep learning model. All six pairwise comparisons (3 hypotheses x 2 baselines) use Bonferroni-corrected significance threshold $\alpha = 0.05/6 = 0.0083$. Wilcoxon signed-rank test, $n = 9,014$ paired test patches.

LIMITATIONS AND FUTURE WORK

- Limitation** CPU-bound training due to on-the-fly skeletonization
- Limitation** Only few model iterations due to long training time
- Limitation** Synthetic gaps may not fully reflect real soil occlusion patterns
- Future** Adversarial discriminator for sharper inpainting (Chen et al. 2019)
- Future** Precompute gap cache to remove CPU bottleneck
- Future** Implementing more loss functions, and even different architectures

REFERENCES

- Chen et al. (2018). Root gap correction with a deep inpainting model. *BMVC*.
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- Sasaki et al. (2017). Joint gap detection and inpainting of line drawings. *CVPR*.